

DUNGEON QUEST

Week 2 Report

Game Design & Planning

Project Title: Dungeon Quest

Project Type: Adventure / Dungeon Exploration Game

Programming Language: Java

Project Duration: 10 Weeks

Week: 2

Introduction

During Week 2, the primary objective was to design the overall structure of the Dungeon Quest game. Instead of implementing gameplay through coding, the focus was placed on planning the game's characters, dungeon environment, level progression, and gameplay mechanics.

This phase serves as the blueprint for future development. By completing all design decisions before implementation, the development process becomes more organized and efficient.

Objectives

The objectives completed during Week 2 are:

- Design the main player character.
- Design enemy characters and define their behaviors.
- Define player abilities and enemy abilities.
- Plan the complete dungeon structure.
- Design multiple game levels with increasing difficulty.
- Plan obstacles and environmental challenges.
- Create the gameplay flow from start to final boss.
- Prepare the Game Design Document (GDD).

Character Design Planning

Character Design Planning defines every character that will appear in the game, their abilities, attributes, and responsibilities.

Main Character Player

Character Name: Souvi

Character Type: Main Hero

Description

Souvi is a brave warrior trapped inside an ancient dungeon. His objective is to escape by defeating monsters, collecting valuable items, solving environmental obstacles, and ultimately defeating the final boss.

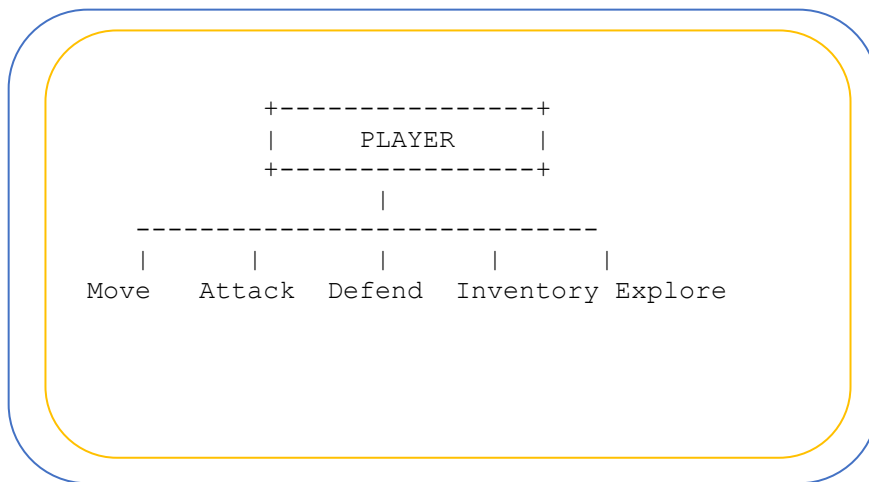
Player Attributes

Attribute	Value
Health	100 HP (Health Point)
Attack	20
Defense	10
Speed	Medium
Weapon	Sword

Player Abilities

- Walk
- Run
- Attack
- Block
- Pick up Items
- Open Treasure Chests
- Collect Keys
- Unlock Doors
- Use Health Potions

Character Diagram



Enemy Character Design

The game contains several enemy types. Each enemy becomes stronger as the player progresses.

Enemy 1 — ZIMBLK

Description

The ZIMBLK is the weakest enemy encountered during the first level.

Attribute	Value
Health	40
Damage	8
Speed	Fast
Difficulty	Easy

Abilities

- Chase Player
- Attack Player
- Patrol Area

Enemy 2 — ASNAKE

Description

ASNAKE guard treasure rooms and attack with swords.

Attribute	Value
Health	70
Damage	15
Difficulty	Medium

Abilities

- Sword Attack
- Guard Treasure
- Defensive Stance

Enemy 3 — RBLIAR

Description

The RBLIAR is stronger and appears in later dungeon levels.

Attribute	Value
Health	120
Damage	22
Difficulty	Hard

Abilities

- Heavy Attack
- Charge Attack
- Area Damage

Final Boss — VERDACHT

Description

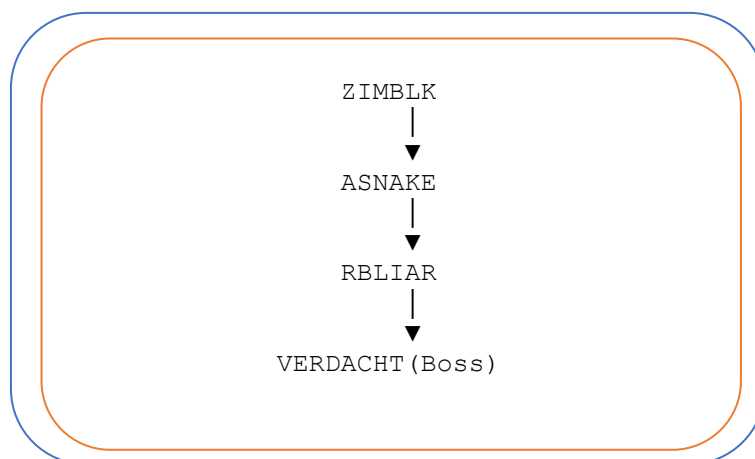
VERDACHT is the **final enemy** guarding the exit of the dungeon.

Attribute	Value
Health	500
Damage	45
Difficulty	Very Hard

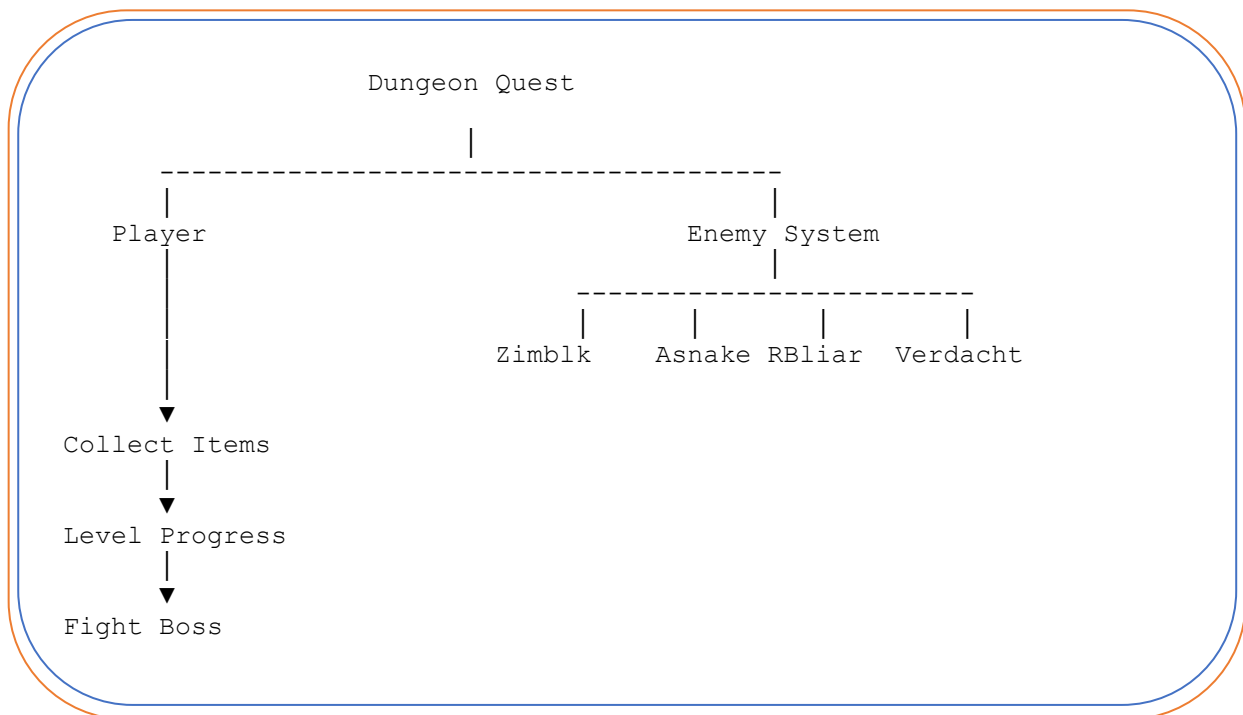
Abilities

- Fire Breath
- Tail Attack
- Flying Attack
- Rage Mode

Enemy Progression Diagram



3.3 Character Relationship Diagram

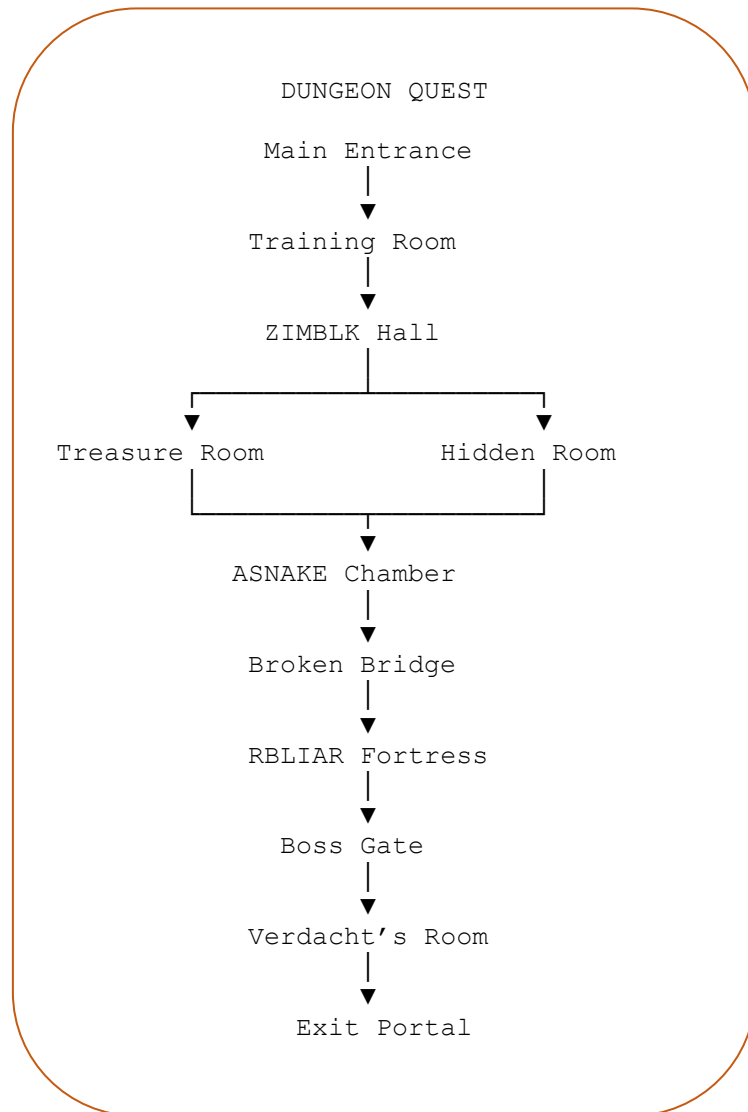


3.4 Character Responsibility Table

Character	Role	Objective
SOUVI	Main Hero	Escape the Dungeon
ZIMBLK	Basic Enemy	Slow down Player
ASNAKE	Mid Enemy	Guard Treasure
RBLIAR	Advanced	Challenge Player
VERDACHT	Final Boss	Protect Dungeon Exit

Dungeon and Level Design

Overall Dungeon Layout:



Level Design

Level 1 – Training Dungeon

Purpose

This level teaches the player the basic controls and gameplay mechanics.

Features

- Easy enemies
- Small dungeon
- Simple pathways
- Health potion tutorial
- Sword tutorial

Enemies

- ZIMBLK

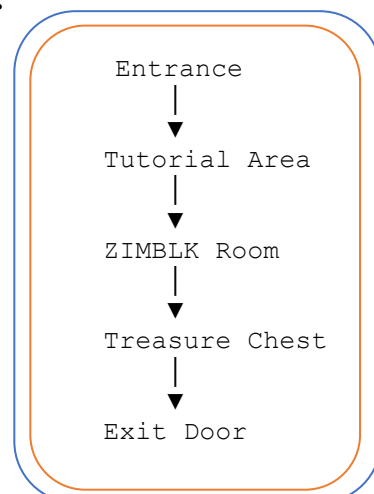
Collectibles

- Small Sword
- Health Potion
- Gold Coins

Objective

Learn movement and defeat the first enemies.

Level 1 Layout:



Level 2 – ASNAKE Cave

Description

The second level introduces stronger enemies and environmental hazards.

Features

- Dark cave
- Locked doors
- Hidden treasure
- Multiple paths

Enemies

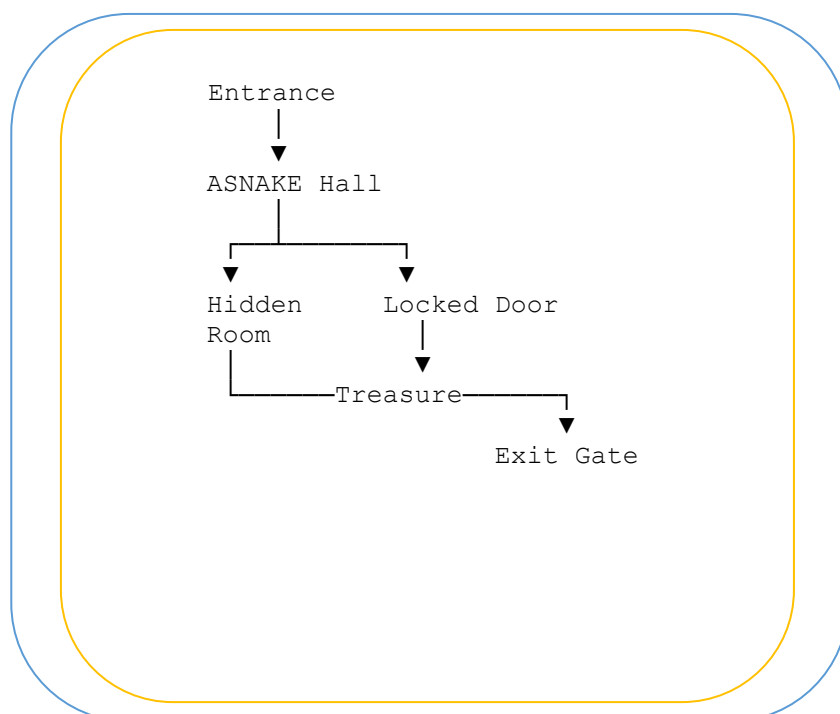
1. ASNAKE

2. ZIMBLK

Collectibles

- Shield
- Dungeon Key
- Healing Potion

Level 2 Layout



Level 3 – RBLIAR Fortress

Description

The player enters the RBLIAR Fortress where stronger enemies patrol every section.

Features

- Large battlefield
- Multiple enemy groups
- Secret passages
- Traps

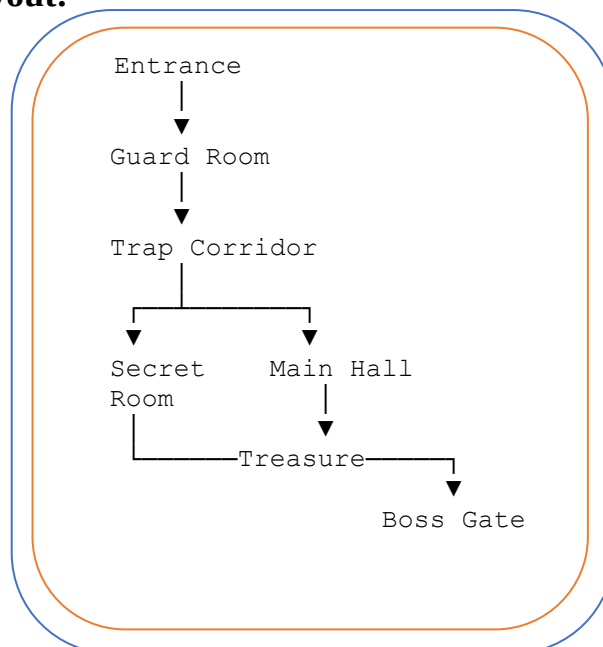
Enemies

- RB Soldiers
- Skeleton Guards

Collectibles

- Magic Sword
- Armor
- Large Health Potion

Level 3 Layout:



Final Level – Dragon King's Chamber

Description

The final level contains the strongest enemy in the game.

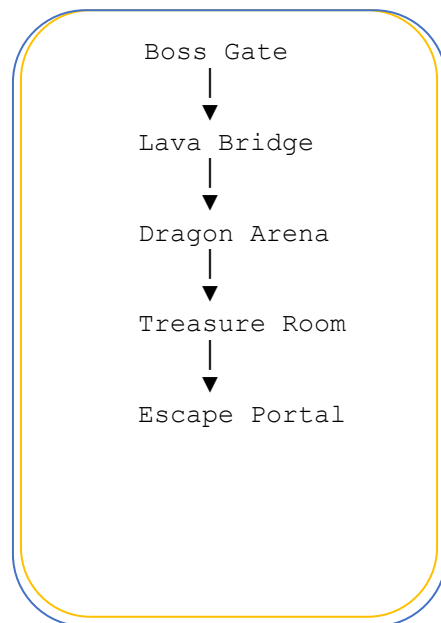
Boss

VERDACHT

Features

- Lava environment
- Fire traps
- Boss arena
- Victory portal

Boss Level Layout



Environment Design

Different environments are planned to make gameplay more engaging.

Area	Description
Stone Corridor	Basic walking path
Treasure Room	Hidden rewards
Cave	Dark exploration area
Lava Area	Dangerous environment
Boss Arena	Final battle location

Obstacles

The player must overcome several obstacles while progressing.

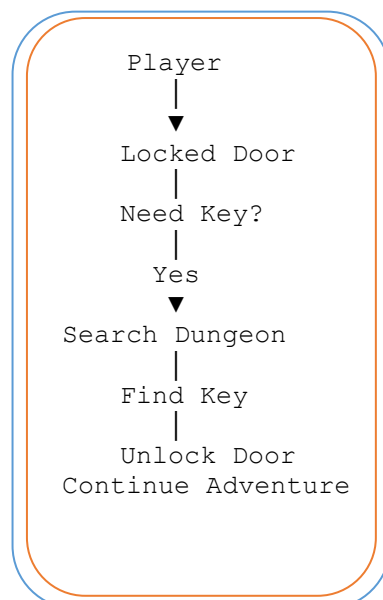
Physical Obstacles

- Locked Doors
- Broken Bridges
- Fire Traps
- Falling Rocks

Puzzle Obstacles

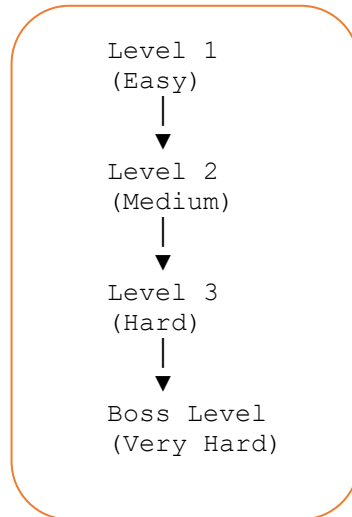
- Find Hidden Keys
- Activate Levers
- Push Stone Blocks
- Open Secret Doors

Obstacle Flow

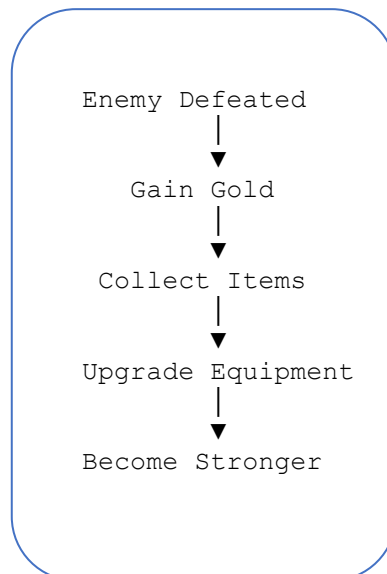


Difficulty Progression

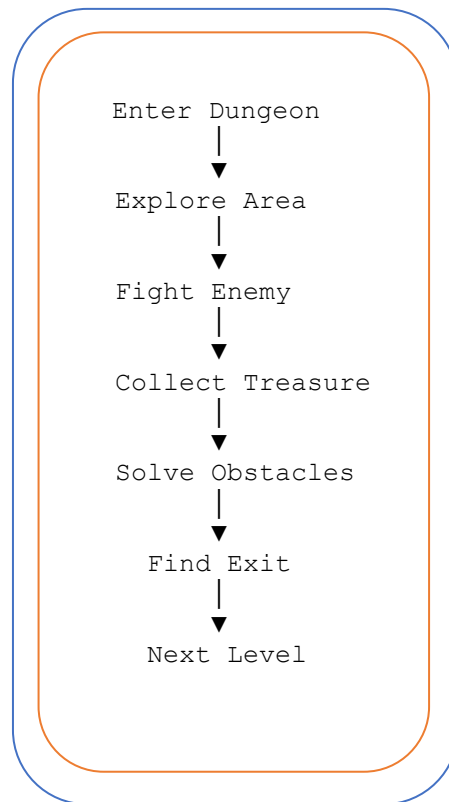
The game's difficulty gradually increases.



Reward Progression

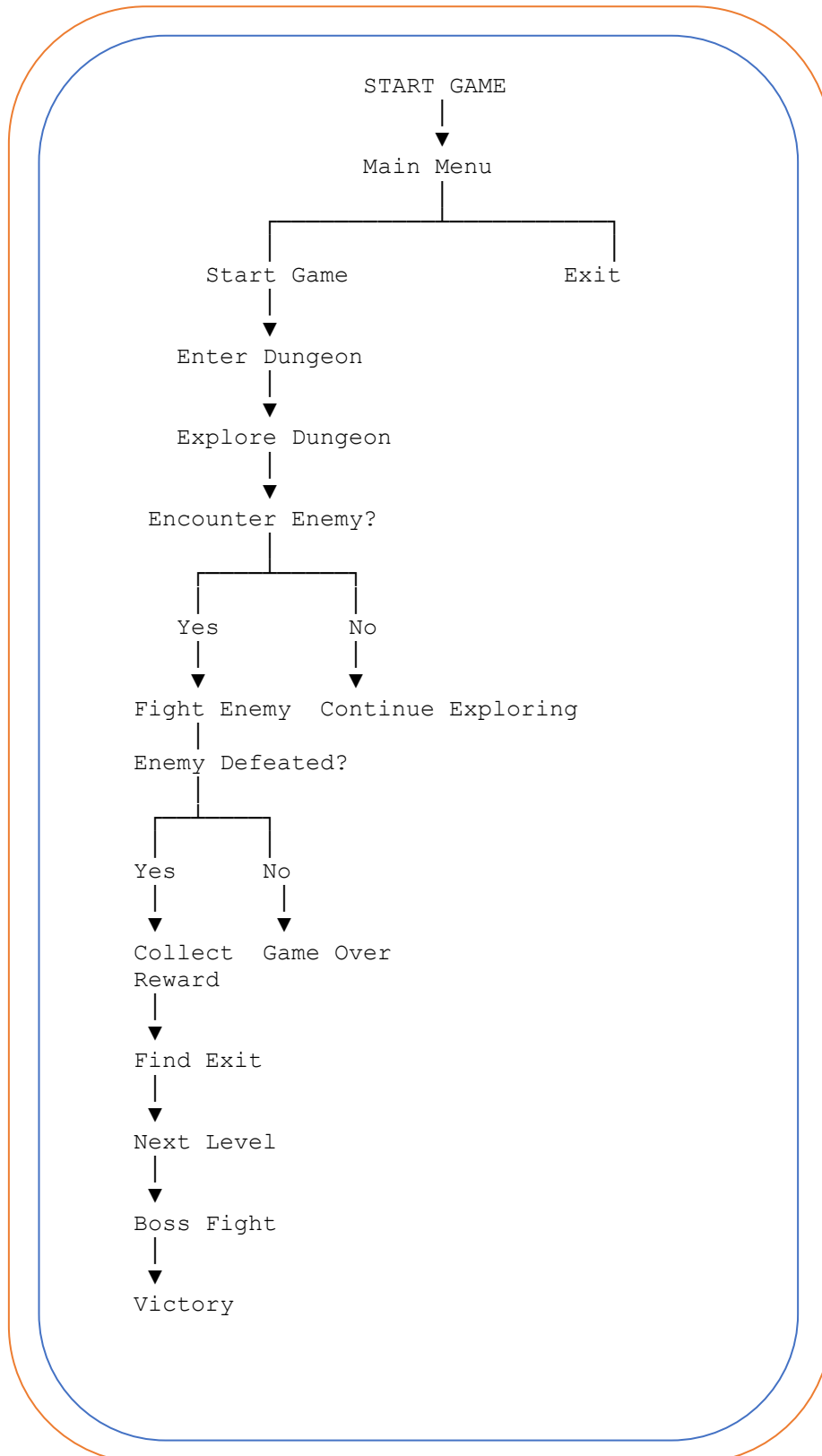


Dungeon Exploration Cycle



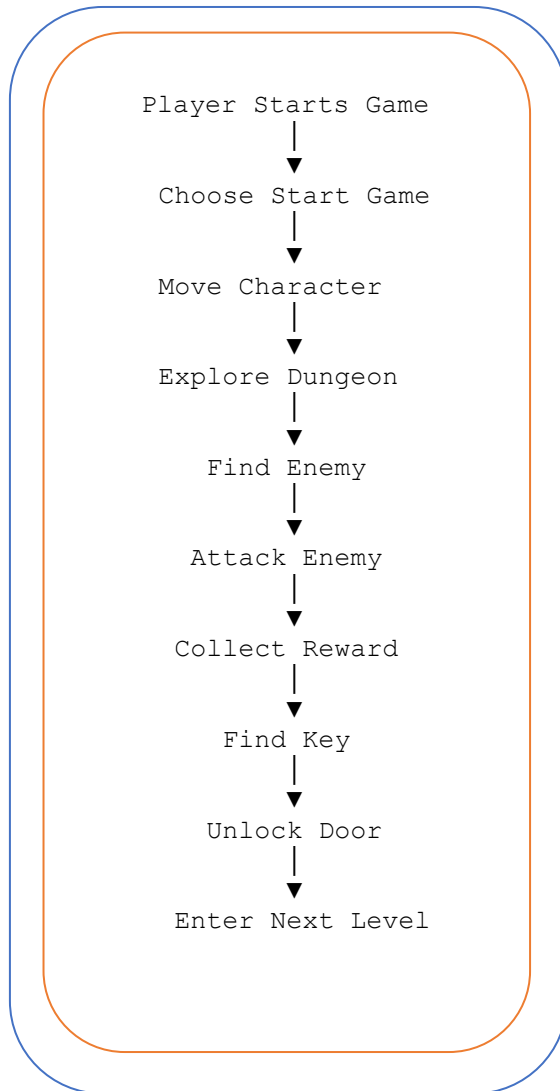
Game Flow Planning

Gameplay Flow Diagram



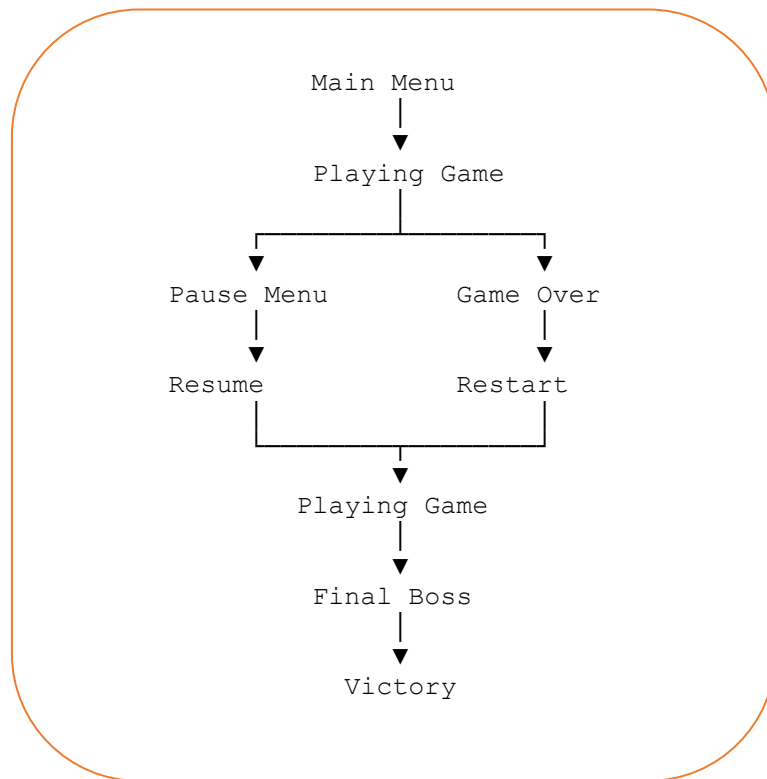
Player Activity Flow

This diagram explains every action the player performs during gameplay.



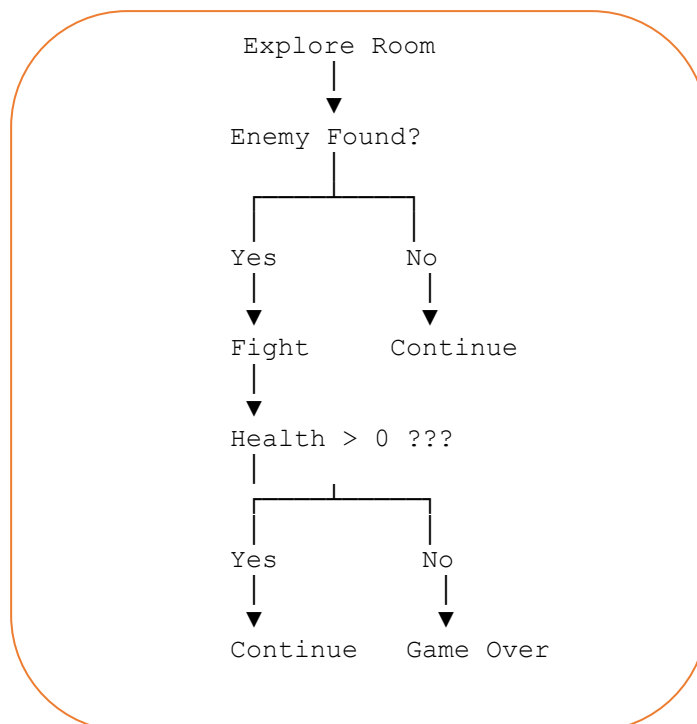
Game State Diagram

The game changes between several different states.



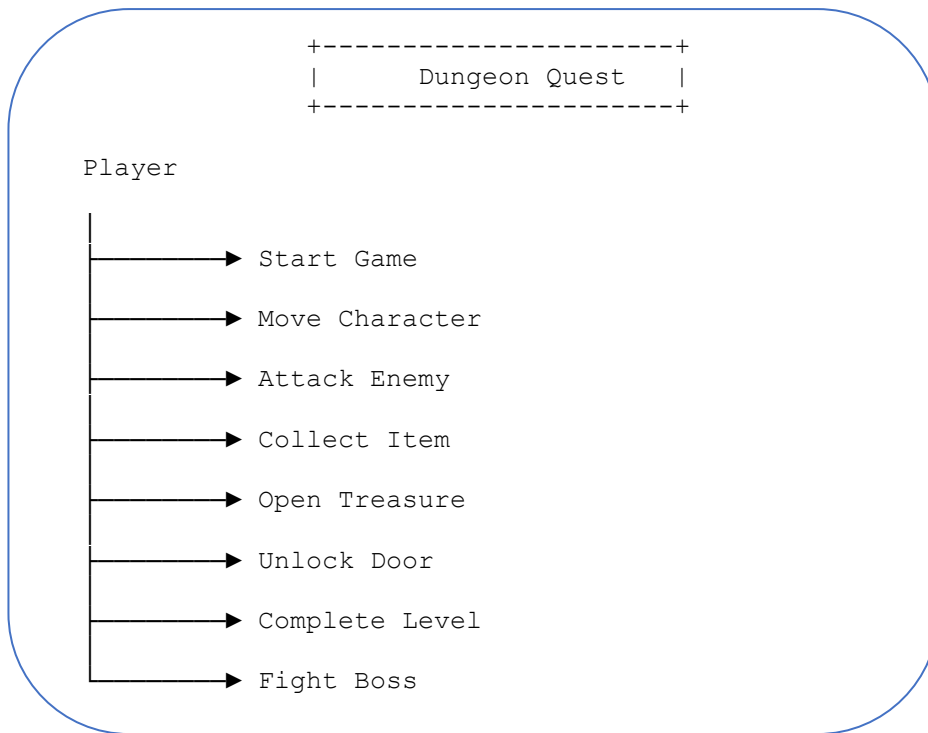
Decision Flowchart

This flowchart represents the decisions the player makes while exploring the dungeon.

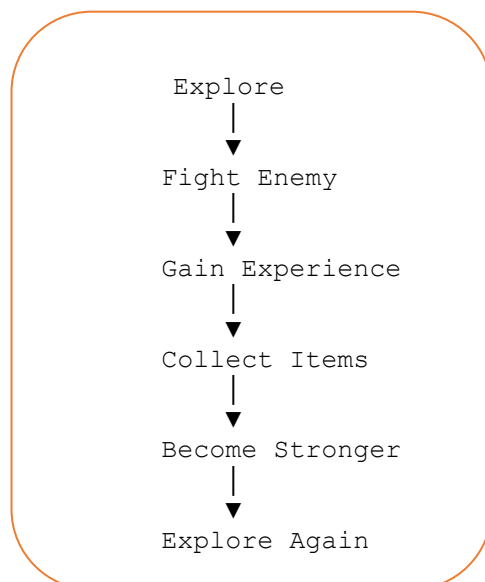


Use Case Diagram

Actors and system functions are identified below.



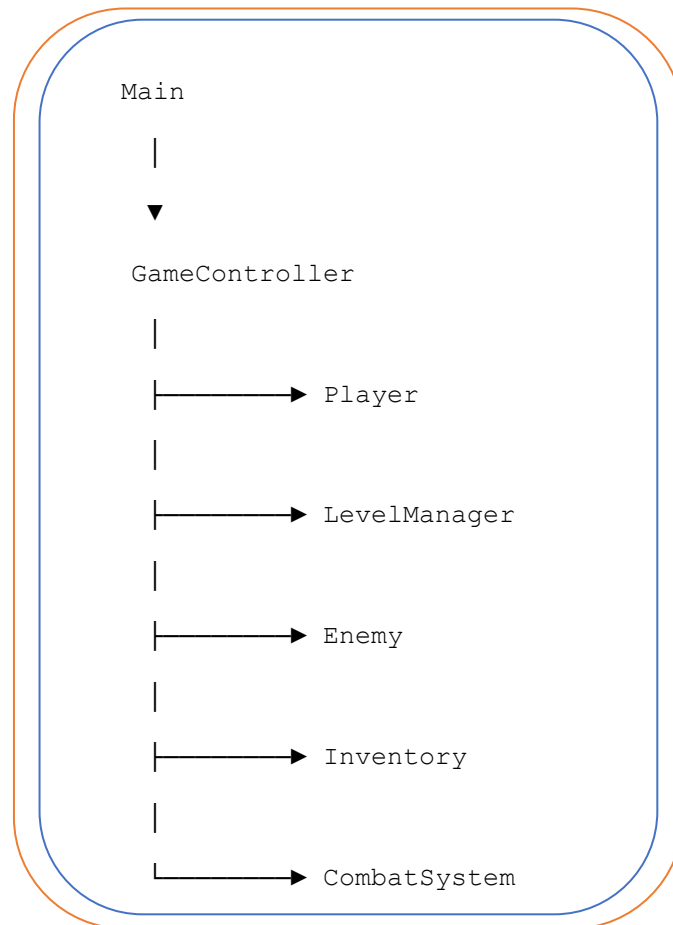
Gameplay Loop



This loop continues until the player reaches the final boss.

Java System Interaction Flow

Although coding has not started, the interaction between Java classes has been planned.



Explanation

- `Main` starts the game.
- `GameController` manages the overall gameplay.
- `Player` receives user input.
- `LevelManager` loads levels.
- `Enemy` controls enemy behavior.
- `CombatSystem` handles battles.
- `Inventory` stores collected items.

Win and Lose Conditions

Win Condition

The player wins if:

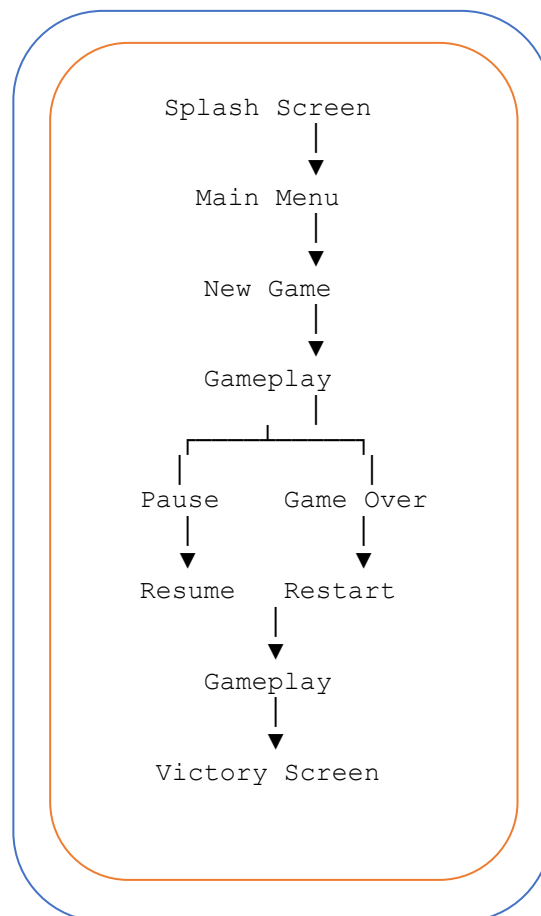
- All dungeon levels are completed.
- The Dragon King (Verdicht) is defeated.
- The player reaches the escape portal.

Lose Condition

The player loses if:

- Player Health reaches **0 HP**.
- The player dies during combat.

Screen Navigation Flow



Game Design Document (GDD)

Project Information of Week 2

Project Name: Dungeon Quest

Project Type: Adventure / Dungeon Exploration

Programming Language: Java

Development Model: Object-Oriented Programming (OOP)

Project Duration: 10 Weeks

Game Description

Dungeon Quest is a Java-based adventure game where the player explores dangerous dungeons, defeats enemies, collects treasures, and progresses through multiple levels before facing the final boss.

The player must survive using combat skills, strategic exploration, and resource management.

Target Audience

Category	Description
Age Group	10+
Game Genre	Adventure, Action
Platform	Desktop (Java Application)
Skill Level	Beginner to Intermediate

Core Gameplay Features

The following gameplay features were finalized during Week 2.

Player System

- Character Movement
- Health Management
- Combat
- Inventory
- Item Collection

Enemy System

- Enemy Patrol
- Enemy Detection
- Enemy Attack

Dungeon System

- Multiple Levels
- Hidden Rooms
- Treasure Chests
- Secret Doors

Inventory System

The player can collect:

- Sword
- Shield
- Health Potion
- Gold Coins
- Dungeon Keys

Combat System

Combat includes:

- Normal Attack
- Heavy Attack
- Defense
- Health Reduction
- Enemy Defeat

User Interface (UI) Design

Main Menu Layout

DUNGEON QUEST

Start Game

Load Game

Settings

Exit

Gameplay Screen Layout

Health Bar:

Gold: 150

Inventory: Sword Potion Key

Player

ZIMBLKASNAKE

Inventory Screen

Inventory

Sword

Shield

Potion

Dungeon Key

Gold

Pause Menu

Game Paused

Resume

Settings

Main Menu

Exit

Victory Screen

Congratulations!

You Escaped the Dungeon

Score: 5800

Play Again

Exit

Game Over Screen

GAME OVER

Health Reached Zero

Restart

Exit
